Menopause

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Abstract. Menopause affects the female body and quality of life. Diet, intermittent fasting, probiotics, and HRT can help manage symptoms and improve well-being.

Published 2025-07-07 · https://microbiomemedicine.com/conditions/menopause · A Microbiome Signatures condition monograph

Overview

Menopause is a natural biological event marking the permanent cessation of menstruation, which occurs after 12 consecutive months without a period, usually between the ages of 45 and 56.[1] This transition results from the loss of ovarian follicles, leading to a decrease in the production of key hormones, particularly estrogen and progesterone.[2] Estrogen's decline impacts numerous bodily systems, leading to a range of symptoms, including hot flashes, night sweats, vaginal dryness, mood swings, sleep disturbances, and cognitive changes.[3] The period before menopause, known as perimenopause, is characterized by irregular menstrual cycles, fluctuating hormone levels, and vasomotor symptoms, which may last for several years. The menopausal transition marks the end of reproductive capacity, but the health implications can persist well beyond menopause, as women can spend up to one-third of their lives in the postmenopausal period.[4] The hormonal changes associated with menopause are well-documented. Still, the broader impacts on various organ systems, including the gut, urinary tract, and skin, are increasingly being explored through research on the microbiome.[5]

Associated Conditions

Menopause is linked to various health conditions that significantly impact women's quality of life.[6] Osteoporosis is one of the most common conditions, as the decline in estrogen accelerates bone resorption, leading to decreased bone mineral density (BMD) and an increased risk of fractures.[7] Cardiovascular disease (CVD) is also a significant concern, as the protective effects of estrogen on the cardiovascular system diminish, increasing the risk of heart disease and stroke.[8] Additionally, musculoskeletal disorders such as osteoarthritis and sarcopenia (muscle loss) are prevalent, contributing to disability and reduced mobility in postmenopausal women.[9] Cognitive decline, particularly Alzheimer's disease, becomes more prominent as estrogen loss may accelerate neurodegenerative processes. Depression is another major issue, often linked to hormonal fluctuations during menopause, while vasomotor symptoms such as hot flashes and night sweats, though distressing, also correlate with an increased risk of cardiovascular events.[10][11] Moreover, sleep disturbances, such as insomnia and sleep apnea, are common, negatively affecting overall health. Lastly, migraines and chronic respiratory diseases like COPD also become more prevalent postmenopause, adding to the health challenges women face during this life stage.[12]

Causes

Menopause is a process that primarily occurs due to the depletion of ovarian follicles, leading to a decrease in estrogen and progesterone production. This decline disrupts the hypothalamic-pituitary-ovarian axis, which regulates the menstrual cycle, and ultimately results in the cessation of menstruation.[13] While the basic cause of menopause, ovarian aging, has been well established, a range of factors can influence the timing and onset of menopause. Genetic predisposition plays a significant role, with research indicating that women whose mothers experienced early menopause are more likely to undergo the transition at a younger age.[14] Environmental and lifestyle factors, such as smoking, body mass index (BMI), physical activity, and stress, are also known to affect the timing of menopause. Smoking, for example, has been consistently linked to earlier menopause, while higher BMI tends to delay its onset.[15][16] Socioeconomic factors, including education level and occupation, have been associated with later menopause, potentially due to differences in healthcare access, nutrition, and overall lifestyle.[17] Hormonal changes, particularly the decline in estrogen, are not the sole contributors to the menopause experience. There is growing evidence suggesting that shifts in the microbiome, particularly in the vaginal and gut microbiota, may also influence the severity of menopause symptoms and contribute to menopause-related health risks.[18] Dysbiosis, or microbial imbalance, resulting from hormonal changes may exacerbate symptoms such as vaginal dryness, urinary tract infections, and gastrointestinal issues. These microbiome shifts, in conjunction with genetic and environmental factors, provide a more complex understanding of menopause beyond just the hormonal decline.[19]

Diagnosis

The diagnosis of menopause is primarily clinical, determined by the cessation of menstrual periods for 12 consecutive months. In women aged 45 and older, menopause is diagnosed based on symptoms alone, without the need for laboratory tests.[20] Hormone levels (FSH, estrogen) are occasionally measured in women aged 40-45, but their reliability is limited due to fluctuations during the perimenopausal period.[21] Women who have undergone hysterectomy may be diagnosed with menopause based solely on the absence of menstrual periods. Microbiome signatures are increasingly being explored as diagnostic tools for menopause-related conditions.[22][23] Studies have investigated microbial changes in the vaginal, gut, and urinary microbiomes during menopause, with a particular focus on dysbiosis as a potential contributor to common symptoms such as vaginal dryness, urinary tract infections, and gastrointestinal issues.[24][25]

Primer

Menopause involves significant hormonal shifts, notably a decline in estrogen, profoundly affecting the vaginal and gut microbiomes.[26] Reduced estrogen decreases beneficial vaginal Lactobacillus populations, increasing pH and susceptibility to infections and genitourinary syndrome symptoms.[27] Similarly, estrogen decline alters gut microbiome composition and diversity, increasing risks for cardiovascular and metabolic diseases.[28] The estrobolome, a subset of gut microbes, is essential in metabolizing estrogen; disruption of this balance during menopause further exacerbates symptoms and associated chronic disease risks.[29] Understanding microbiome-estrogen interactions could thus inform better menopause symptom management and long-term health strategies.

Metallomic Signatures

Exposure to heavy metals such as arsenic, cadmium, lead, and mercury has been increasingly recognized as an important factor influencing menopausal timing and hormonal profiles in midlife women. High levels of these metals are associated with fertility disorders, endocrine dysregulation, and earlier onset of menopause.[30] In particular, arsenic and lead have been linked with significantly earlier natural menopause; for instance, higher urinary concentrations of arsenic and lead have been associated with menopause occurring up to nearly one year earlier compared to women with lower exposures. These metals can disrupt sex hormone levels by reducing circulating estradiol (E2) and increasing follicle-stimulating hormone (FSH), which contributes to ovarian aging.[31] Cadmium and mercury similarly alter hormone levels, with cadmium notably associated with increased sex hormone-binding globulin (SHBG) concentrations, influencing free hormone availability.[32] Heavy metals have been found to interact with menopausal status and exacerbate health risks, including metabolic disturbances and cardiovascular conditions. Collectively, these suggest that heavy metal exposure during midlife significantly contributes to menopause-related endocrine and metabolic disruptions, underscoring the importance of assessing and mitigating environmental metal exposure in menopausal women.[33]

Metabolomic Signatures

Menopause induces substantial shifts in the metabolome, which profoundly influences women's metabolic health. Estrogen significantly regulates lipid and amino acid metabolism, and its reduction during menopause results in proatherogenic lipid alterations, including increased concentrations of LDL cholesterol, very-low-density lipoprotein triglycerides, and apolipoprotein B (apoB), which elevates cardiovascular disease risks.[34][35] Estrogen deficiency is also implicated in disturbed fatty acid metabolism, including elevated circulating levels of saturated and monounsaturated fatty acids, as well as specific changes in omega-7 and omega-9 fatty acid profiles, potentially exacerbating metabolic syndrome and insulin resistance in postmenopausal women.[36] Importantly, estrogen modulates the metabolism of charged metabolites, particularly amino acids; menopausal transition has been linked to increased levels of branched-chain amino acids like leucine and isoleucine, which are known markers for insulin resistance, obesity, and cardiovascular conditions. Menopause-related estrogen decline disrupts the metabolic pathways involving the tricarboxylic acid (TCA) cycle and urea cycle, leading to altered citrate, glycine, and homocysteine metabolism.[37] These metabolic shifts underscore estrogen's critical role in maintaining systemic metabolic homeostasis and highlight how its deficiency during menopause contributes significantly to increased risks for metabolic disorders, cardiovascular disease, and osteoporosis in postmenopausal women.

Interventions

Microbiome-targeted interventions (MBTIs) for menopause focus on correcting microbial imbalances linked to menopausal symptoms and hormonal shifts. By addressing dysbiosis in the gut, vaginal, and urinary microbiomes, these interventions aim to alleviate common menopause symptoms like hot flashes, mood changes, and metabolic disruptions. MBTIs are gaining attention as a potential therapeutic approach to improve overall health and manage menopause-related conditions, offering a new, microbiome-centered strategy for enhancing women's well-being during this transition.

Interventions	Mechanisms of Action	MBTI Status
Prebiotic Supplementation	Prebiotics enhance the growth of estrogen-metabolizing bacteria like *Bifidobacterium* and *Lactobacillus*, improving estrogen metabolism and alleviating menopausal symptoms like hot flashes.[38]	Promising Candidate
Ketogenic Diet	Alters gut microbiome composition by promoting beneficial microbes, improving insulin sensitivity and reducing inflammation, which are critical for managing menopausal symptoms.[39]	Promising Candidate
Intermittent Fasting (IF)	IF enhances gut microbiome diversity, reducing inflammation and improving metabolic health, which could alleviate metabolic disturbances in postmenopausal women.[40][41]	Under Investigation
Phytoestrogens	By improving estrogen balance, phytoestrogens have been shown to reduce the frequency and severity of hot flashes and other menopausal symptoms.[42]	Experimental
Dietary Fiber and Polyphenols	Soy isoflavones, through dietary fibers and fermented soy products, improve gut health, estrogen metabolism, and cardiovascular health in postmenopausal women.[43][44]	Experimental
Probiotic Supplementation	Probiotics such as *Lactobacillus* strains help regulate vaginal and intestinal microbiota, improving immunity and alleviating menopausal symptoms, particularly vaginal dryness and infections.[45][46]	Under Investigation
Combination of Black Cohosh and Rhodiola	The combination of *Actaea racemosa* (Black Cohosh) and *Rhodiola rosea* improves menopausal symptom relief, including hot flashes and psychological symptoms like anxiety.[47][48][49]	Promising Candidate
Menopausal Hormone Therapy (MHT)	Hormonal therapy using estrogen and progestogen aims to alleviate common menopausal symptoms such as hot flashes, sleep disturbances, and prevent bone loss. It may also improve cardiovascular health.[50][51]	Validated
Vaginal Microbiota Transplantation (VMT)	VMT involves transplanting vaginal microbiota to restore a healthy microbial environment, potentially reducing infections and alleviating vaginal dryness and irritation associated with menopause.[52]	Experimental

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